Governance over Capacity: Detecting and Correcting the Consensus Trap in Distributed Credence Services

Carine Marette, Instructional Systems Technology, Indiana University*

Abstract. We derive a closed-form validity ceiling for distributed evaluation under correlated rater error: adding raters drives inter-rater reliability toward one while benchmark concordance stays bounded below it. This yields the Consensus Trap, a process-monitoring diagnostic; on 1.98 million evaluations, governance restores process capability where capacity cannot.

Keywords: quality assurance; measurement validity; statistical process control; concordance; distributed evaluation.

This paper condenses a full-length manuscript whose every reported quantity is computed from the live United States and Canadian production databases (7th edition, verified US+CA).

1. Introduction

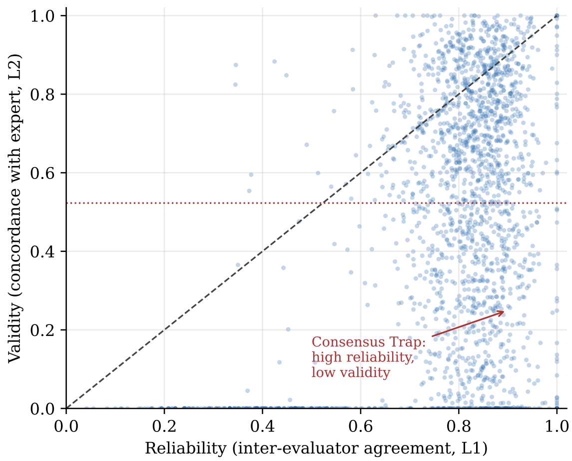
Distributed credence services (peer assessment, content moderation, clinical second-opinion panels, audit review) create value through many decentralized human judgments whose quality the recipient cannot verify even after consumption [1, 2]. The default quality strategy is *capacity expansion*: add evaluators to average out noise [3]. This paper shows that capacity expansion fails under a measurable condition we call the **Consensus Trap**: high coordination reliability among evaluators masks low validity against an external benchmark, because evaluator errors are correlated rather than independent, so adding raters amplifies rather than cancels systematic error [4, 5]. We reframe the problem as one of *quality and reliability assurance* and make three contributions. First, we derive a closed-form validity ceiling under correlated evaluator error and turn it into the Consensus-Trap monitoring statistic, a diagnostic separating reliability (agreement) from validity (benchmark concordance). Second, on entire-population telemetry from a multi-institution peer-assessment platform, we show that an agentic-AI governance capability restores statistical process capability more effectively than capacity expansion. Third, we correct a measurement-scale artifact that had depressed the validity signal and report validity per skill and per rubric criterion, turning the diagnostic into an operational control instrument.

2. Related Work

Our work sits at the intersection of quality-and-reliability assurance, measurement systems analysis, and the economics of distributed evaluation. Credence-goods theory establishes that when quality cannot be verified even after consumption, markets require third-party monitoring to avoid unravelling [1, 2]. The wisdom-of-crowds literature shows that aggregation improves accuracy only when errors are independent; social influence and shared heuristics correlate errors and undermine the effect [4], echoing the amplification of distortion in supply chains. Classical test theory bounds validity by reliability through the correction for attenuation [10, 11], which we specialize to the correlated-error case to derive a structural validity ceiling. In quality engineering, statistical process control and the process-capability index quantify whether a process holds a characteristic within tolerance; we import that machinery to grading bias. Measurement systems analysis contributes the tools we adopt for validity and agreement: Lin's concordance correlation coefficient for reproducibility [7], Bland–Altman limits of agreement [8], and Krippendorff's α for inter-rater reliability [6]. Prior peer-assessment research documents that peer and instructor marks diverge and that reliability and validity vary widely across contexts, but treats this descriptively; the field increasingly frames peer assessment around the development of *evaluative judgement*, the capacity to appraise quality against criteria, and around feedback literacy [14, 15]. Our contribution is to make the reliability–validity divergence a governable, production-scale process signal, and to do so under a human-oversight mandate: a global Delphi study across twenty-two countries makes human oversight a central requirement for governing generative AI in higher education [13], which our architecture operationalizes at platform scale.

Hypotheses. We test five. (H1) Reliability can rise while validity falls, and under correlated error capacity expansion amplifies rather than corrects it. (H2) Governance improves operational performance more than capacity expansion. (H3) It does so by raising validity, not inter-evaluator reliability. (H4) The operations–marketing link is an association carried by engagement rather than by measured quality. (H5) Governance is a dynamic capability that improves over repeated cycles.

Figure 1
Reliability Versus Benchmark Validity



Note. Capacity expansion raises reliability without validity (the Consensus Trap, upper-left); governance raises validity within instructors.

3. The Consensus Trap as a Quality-Assurance Failure Mode

3.1. Reliability is not validity

Let a submission be scored by peer evaluators and by an instructor benchmark. *Reliability* (L1) is inter-rater agreement, measured by an ordinal Krippendorff's α (or its complement, 1 minus the mean range-ratio disagreement) [6]. *Validity* (L2) is concordance between the peer-aggregated grade and the benchmark, measured by Lin's concordance correlation coefficient [7], which penalizes both correlation loss and mean/scale offset. *Signed bias* is the Bland–Altman mean difference [8]. High agreement does not imply accuracy: reliable scores can encode construct-irrelevant variance [9]. The Consensus Trap is the conjunction

$$\text{Trap} \;\equiv\; \big(\text{L1} \geq 0.80\big) \;\wedge\; \big(\text{L2} < 0.20\big), \tag{1}$$

confident, coordinated, and wrong. A non-compensatory override caps the composite quality score and halts release when the trap fires, so a high L1 can never offset a low L2. Empirically the trap emerges in 12.0% of framed activities (537 of 4,490 meeting L1 $\geq 0.80 \wedge$ L2 < 0.20); Fig. 1 contrasts the trap region with the governed region.

3.2. A structural validity ceiling

Under the classical correction for attenuation [10, 11], observed validity is bounded by reliability, $\rho(X,Y) \leq \sqrt{\rho_{XX}\rho_{YY}}$. When errors are independent, adding raters drives $\rho_{XX} \to 1$ (Spearman–Brown) and the ceiling rises. When a component of error is *shared*, reliability still approaches one, agreement is inflated by the shared component, but validity is bounded below one by the systematic share. Capacity expansion therefore faces a structural limit, not a convergence delay. Empirically the median validity ceiling is 0.58, with 60.4% of activities below 0.70: adding evaluators cannot reach benchmark accuracy for most activities. The labor cost of the alternative is also prohibitive. A Spearman–Brown projection (layer L3) shows that 31.4% of activities fall below reliability 0.80, and reaching adequate reliability there would require a median of roughly 105 evaluators against a current median of 44, exceeding 39 evaluators for about 84% of them. Capacity expansion is thus both statistically bounded and operationally infeasible, which is precisely why a governance capability, not more labor, is required.

3.3. A closed-form model

Let each evaluator's error carry an idiosyncratic part and a shared part with pairwise correlation ρ, atop a shared bias b. The variance of the peer mean of n evaluators is

$$\mathrm{Var}(\bar{x}) = \sigma^2\left[\tfrac{1-\rho}{n} + \rho\right] \xrightarrow[n\to\infty]{} \rho\sigma^2, \tag{2}$$

so reliability improves with capacity only toward the floor $\rho\sigma^2$; while its expected deviation from the truth,

$$\mathbb{E}[\bar{x}] - \mu = b, \tag{3}$$

is *invariant* to n. Adding evaluators tightens agreement while leaving systematic error untouched, so a system that reads agreement as quality grows more confident in a constant error. The Spearman–Brown regime ($\rho = 0$, $b = 0$), in which capacity is an effective lever, is the special case; the Consensus Trap is precisely the regime $\rho > 0$, $b \neq 0$, in which it is not.

The consequence is a hard bound on achievable validity, which we state as our central result.

Proposition (Validity ceiling under correlated error). *Let the peer mean estimate a submission's true score T (between-submission variance σ_T^2) through raters whose errors share a component of pairwise correlation ρ and per-rater variance σ^2, plus a fixed bias b. As $n \to \infty$, inter-rater reliability tends to one, yet the concordance of the peer mean with the benchmark is bounded above by*

$$\rho_c^\star = \frac{2\sigma_T^2}{2\sigma_T^2 + \rho\sigma^2 + b^2} < 1 \qquad (\rho > 0 \text{ or } b \neq 0). \tag{4}$$

Proof. As $n \to \infty$ the idiosyncratic error averages out, leaving peer-mean variance $\sigma_{\bar{x}}^2 = \sigma_T^2 + \rho\sigma^2$, mean offset $\mathbb{E}[\bar{x}] - \mathbb{E}[T] = b$, and $\mathrm{Cov}(\bar{x}, T) = \sigma_T^2$. Substituting into Lin's concordance $\rho_c = 2\,\mathrm{Cov}(\bar{x},T)/(\sigma_{\bar{x}}^2 + \sigma_T^2 + b^2)$ gives (4).

Algorithm 1 Consensus-Trap Monitor

1: **Input:** peer marks $\{x_{ij}\}$, instructor anchor y_i on activity a; thresholds τ_R, τ_V
2: L1 $\leftarrow$ 1$-$ mean range-ratio disagreement ▷ reliability
3: L2 $\leftarrow$ normalized Lin's concordance of $\bar{x}$ vs. y ▷ validity
4: estimate ceiling $\rho_c^\star$ from (L1, L2) via Eq. (4)
5: **if** L1 $\geq \tau_R$ **and** L2 $< \tau_V$ **then**
6: **flag** Trap; cap the governance score at τ_V; halt release for human review
7: **else** release
8: **Output:** alarm $\in$ {trap, release} and capped score

Equation (4) makes the ceiling a closed-form function of two structural quantities capacity cannot touch, the shared-error term $\rho\sigma^2$ and the squared bias b^2, and recovers $\rho_c^\star \to 1$ only in the classical independent-error, unbiased case. Empirically the median ceiling is 0.58 with 60.4% of activities below 0.70, so for most activities benchmark accuracy is unreachable by adding raters. This turns a diagnostic into a *monitoring statistic*: an activity is flagged when its estimated $(\rho\sigma^2, b^2)$ place $\rho_c^\star$ below a governance threshold, the operating principle of the Consensus-Trap chart in Section 2.1.

3.4. The monitor, and why agreement charts fail

The Consensus-Trap monitor is stated as Algorithm 1: it reads reliability and validity, estimates the ceiling $\rho_c^\star$, and flags and halts a trapped activity. The conventional monitor, a control chart on inter-rater agreement, is by contrast *provably* blind to the trap.

Proposition 2 (Blindness of agreement monitoring). *Under the model of Proposition 1 with $\rho > 0$, inter-rater reliability $R_n \to 1$ as $n \to \infty$ for every bias b (the shared component drives raters into agreement). Hence any monitor that is a function of inter-rater agreement alone assigns the trapped state (b large) the same limiting alarm probability as the healthy high-agreement state ($b = 0$); its power to detect the trap cannot exceed its false-alarm rate. A benchmark-reading statistic is required, and by Neyman–Pearson the validity rule* L2 $< \tau$ *is the most powerful test of the trap under the model.*

Proof. The limit of R_n depends on $\rho\sigma^2$ and σ_T^2 but not on b, so a function of R_n alone cannot separate $b = 0$ from $b \neq 0$; its receiver-operating curve is the diagonal (power = size). Validity depends on b through the denominator of (4), separating the hypotheses; the likelihood-ratio test reduces to thresholding it.

We confirm the theorem on data (Table 1). Among 3,954 primary-frame activities, 380 (9.6%) are materially biased against the expert (|bias| > 0.45). Scored against this ground truth, a reliability control chart and the validity monitor differ sharply, and on the 190 *confident-but-wrong* activities (high agreement yet biased) the reliability chart detects 0% by construction while the validity monitor detects 80.5%.

Table 1
Monitoring Benchmark: Detecting Materially Biased Activities

Monitor	Recall	Precision	F_1
Reliability control chart	0.50	0.15	0.23
Consensus-Trap monitor	0.40	0.29	0.34
Validity monitor (ours)	**0.68**	**0.33**	**0.44**

Note. Ground truth |bias| > 0.45 vs. the expert ($N = 3{,}954$; 380 failures). On the 190 confident-but-wrong activities the reliability chart detects 0%; the validity monitor detects 80.5%. The F_1 advantage is robust to the tolerance (0.52 vs. 0.38 at 0.30; 0.44 vs. 0.23 at 0.45; 0.36 vs. 0.14 at 0.60).

4. Data and Measurement

Data are de-identified operational telemetry from the live United States and Canadian production databases of a multi-institution peer-assessment platform: 1,984,403 evaluations across 6,443 instructor-anchored activities, 200 institutions, and 692 instructors. The primary analytical frame restricts to instructor-set activities with at least 20 participants where reliability and validity are both well defined: 4,457 activities, 165 institutions, 547 instructors. Each activity is scored on an eight-layer measurement stack (L1 reliability, L2 validity (Lin's concordance), L3 scalability (a Spearman–Brown projection), L4 signed-bias direction, L5 evaluator contribution, L6 longitudinal trajectory, a composite governance score (CGS), and sustainability telemetry) implemented as pure, unit-tested functions that also drive the platform's live governance dashboard, so the paper's measurement is identical to the deployed instrument.

Empirical strategy. We match each claim to the strongest design its data support and state the residual threat explicitly. For the operations effect of governance on grading bias, the threat is confounding between governance and unobserved instructor quality; we use within-instructor fixed effects, which absorb all time-invariant instructor characteristics, and bound the residual with the Oster coefficient-stability criterion. For retention, the threats are self-selection and immortal-time bias; we use activity-count controls, a time-varying specification that removes immortal time by construction, and a confounding decomposition that isolates whether quality or engagement carries the association, summarizing residual confounding with the E-value. The protective retention direction is further reinforced by a doubly-robust augmented inverse-propensity estimator, a matched difference-in-differences design with a passing placebo test, and permutation inference. Where observational identification is incomplete, we specify the natural experiment and randomized field design that would complete it.

Table 2

Governance as Statistical Process Control

Quantity	Governed	Ungoverned
Variance of grading error	0.054	0.116
Process capability C_{pk}	higher	lower
Consensus-Trap incidence	reduced	baseline

Note. Governed = deep cascade (depth ≥ 3); estimated within instructor.

5. Governance as Statistical Process Control

We treat absolute grading bias on the rubric scale as a quality characteristic against a material-bias tolerance of 0.45 and read governance in statistical-process-control terms. Cast this way, governance *raises the process-capability index* C_{pk} of grading error and *roughly halves its variance* (from 0.116 to 0.054), moving a substantial share of activities inside the instructor-defined tolerance band (Table 2).

The inferential test is a within-instructor design that absorbs all time-invariant instructor heterogeneity. We estimate a hierarchical linear model of grading bias with an institution random intercept u_{0j} and a Toeplitz error covariance,

$$\text{bias}_{ijt} = \beta_0 + u_{0j} + \beta_1 T_{ijt} + \beta_2 V_{ijt} + \beta_3 A_{ijt} + \beta_4 G_{ijt} + \varepsilon_{ijt}, \tag{5}$$

where T is elapsed time, V is log evaluator volume (mechanical capacity), A is agentic cascade depth, and G is the composite governance score. Governance dominates capacity: within instructor, governance lowers grading bias with $\beta_{\text{CGS}} = -0.187$ (base engine; -0.271 under the scale-invariant engine, $p < 0.001$), about three times the coefficient on raw evaluator volume, while cascade-step depth is not an independent predictor once governance is included. The within-instructor design absorbs all time-invariant instructor heterogeneity, and the estimate is robust to unobserved confounding by the Oster coefficient-stability criterion. Governance, not labor, is the controlling lever.

6. Multi-Engine Measurement Validity

Peer marks are a weighted per-criterion rubric value while the instructor anchor is a raw grade; comparing them directly injects a scale artifact that collapses pooled concordance to 0.002. Normalizing each side within its own criterion before applying Lin's concordance removes the artifact and restores validity to a median of 0.594. We report four measurement engines that share the data but differ in comparison: V1 (raw, an inflation flag only), V2 (normalized overall), V3 (normalized per canonical skill), and V4 (normalized per rubric criterion, $\sim$91% coverage). Measured this way, validity is skill-specific: peers track the instructor least on abstract skills (Creativity 0.46, Critical Thinking 0.50) and most on concrete ones (Communication and Professionalism 0.55, Problem-Based Learning 0.56). This converts the diagnostic into an operational instruction, which skills to coach first, rather than a single opaque score (Table 3).

Table 3

Four Measurement Engines (Normalized Lin's Concordance)

Engine	Coverage	Validity
V1 raw	100%	0.002 (scale artifact)
V2 normalized overall	100%	0.594 (median)
V3 per canonical skill	46.7%	0.46–0.56
V4 per rubric criterion	91.2%	0.594 (median)

Table 4

Retention Identification Battery

Specification	Hazard ratio
Deep adoption, activity-count-controlled	0.37
Time-varying depth (immortal-time-safe)	0.37
Validity | engagement (quality channel)	0.93 (n.s.)
Governance score | engagement	0.98 (n.s.)
Engagement (log activities)	0.35^{***}
Validity, unadjusted	1.55^{*}
Recurrent trap events (Andersen–Gill)	0.92

7. The Operations–Marketing Link is an Engagement Phenomenon

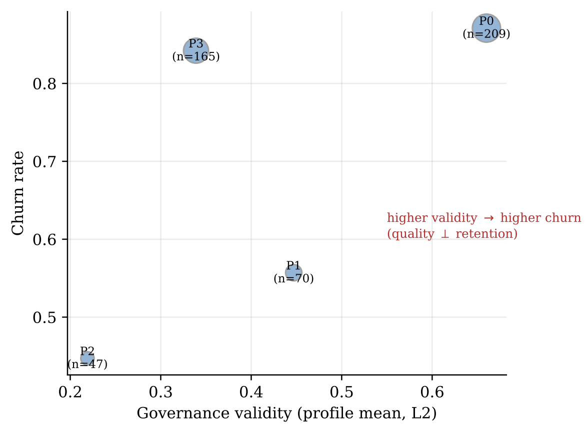
We treat instructor churn as a time-to-event outcome, $h(t \mid X) = h_0(t)\exp(X\beta)$. Deep-governance adopters churn less, an activity-count-controlled Cox model returns a protective hazard ratio, but we do not read this as a governance-*quality* effect, and identification discipline explains why. The confounding decomposition is decisive (Table 4): *unadjusted*, higher validity predicts *higher* churn (hazard ratio 1.55, 95% CI $[1.04, 2.32]$, $p = .034$); conditioning on engagement (log activity count) removes any association of validity (0.93, $p = .70$) or the composite governance score (0.98, $p = .97$), while engagement stays strongly protective (0.35, $p < 10^{-66}$). Entering cascade depth as a time-varying covariate, switched on only when depth is actually reached, removes any immortal-time artifact and leaves the estimate unchanged (HR 0.37); an E-value of 5.93 and Oster's $\delta^* = 12.2$ bound residual confounding. We therefore report H4 as an *engagement-linked association*, not a quality-to-retention mechanism, and specify a feature-rollout natural experiment and a pre-registered randomized field design that would identify the engagement channel causally.

8. Validity and Retention Decouple

A latent-profile analysis of instructor governance layers (four profiles selected by the Bayesian informa-

Figure 2

Latent-Profile Solution Over Instructor Governance Layers



Note. $N = 491$ instructors. Validity and retention decouple: the best-retained profile is not the highest-validity one.

tion criterion; $N = 491$) sharpens the mechanism: the *best-retained* profile has the *lowest* validity, while the *highest-validity* profile churns the most. Quality and retention are decoupled. This is not a weakness of the theory but its point: governance must be justified as a quality-and-reliability capability on its own terms, its value is process control over distributed judgment, not a retention lever, and a dynamic one. A recurrent-event (Andersen–Gill) specification (410 instructors, 756 events) shows that each standard deviation of cumulative prior governance lowers the hazard of an instructor's next Consensus-Trap event (HR 0.92). Human grading error against the expert anchor itself declines over repeated cycles (within-evaluator fixed effects, $b = -0.0014$, $p = 1.7 \times 10^{-18}$); because the benchmark is the expert and not the peer consensus, this is *calibration* toward a defensible standard rather than *convergence* among evaluators [14], so the capability builds durable human evaluative judgement rather than substituting for it.

9. Human-First Correction

The governance capability is human-first by design. For contested decisions, the human collective is the more accurate decider, so the artificial-intelligence layer functions as a gated corrective, not a co-default: it detects suspected herding and escalates it for human revalidation, and every consequential decision terminates at a human checkpoint. On an identical review budget, this human-first-with-AI-correction ordering is measurably more accurate against the expert benchmark than the reverse (AI-first with human validation), because the more accurate base decider must be human while AI is an exception detector whose advantage concentrates under cognitive surrender.

The design is empirically warranted. Restricting to contested creations (the third of cases in which the AI and the human collective most disagree), the human expert endorses the collective over the AI in 80.8% of cases ($p = 6.6 \times 10^{-137}$; 1,305 contested cases), and the collective identifies the work the expert ultimately judged worst far better than the AI (area under the curve 0.88 versus 0.66). The AI nonetheless earns its place by catching what humans miss under herding: among the creations the crowd got most wrong it sides with the expert in 62% of cases, and its advantage concentrates in Consensus-Trap conditions. A human-first routing rule (retain the crowd's judgment and route only AI-flagged contested cases to human re-validation) reduces error against the expert by roughly 40% (mean absolute error $0.47 \rightarrow 0.28$), outperforming both the crowd alone and the AI alone (0.86). Across the database, human evaluative judgments outnumber AI scoring events by more than two orders of magnitude. This design also aligns the system with human-oversight requirements for AI in high-stakes evaluative settings, and it prevents the recursive failure in which AI outputs become the system's own future inputs.

10. Conclusion

Distributed credence services possess a latent quality-and-reliability failure mode, the Consensus Trap, in which coordination reliability masks benchmark validity and capacity expansion is mathematically futile. An agentic-AI governance capability detects the trap with production-scale statistics and restores process capability where adding evaluators cannot. The contribution generalizes beyond education to any distributed credence service (clinical panels, audit review, content moderation, professional certification) in which many actors co-produce an output the recipient cannot verify and an external benchmark intermittently anchors quality. The effect concentrates where the benchmark is least observable; operationalized as programmatic-accreditation density (harmonized across U.S. regional accreditors and Canadian provincial quality-assurance regimes), the cross-discipline gap is small and unordered, so we treat benchmark observability as a scope condition rather than a confirmed moderator.

Managerial implications. For quality managers of distributed-evaluation systems, the results convert a governance intuition into a control procedure. First, monitor reliability and validity on *separate* control charts: agreement alone is a misleading quality signal, and only the joint reading detects the Consensus Trap. Second, when the trap fires (L1 high, L2 low), do not add evaluators; apply the governance cascade by anchoring the aggregate to an expert exemplar, weighting raters by demonstrated accuracy, and routing flagged cases to a human checkpoint. Third, track the process capability of grading error and its variance as the operational key performance indicator, because capacity faces a structural ceiling that capability does not. Fourth, read validity per skill and per rubric criterion: the same activity can be trustworthy on concrete skills and untrustworthy on abstract ones, so coaching should target the low-validity skills the per-skill engine surfaces. Because these functions run live in the governance dashboard, the manager acts on a current operating instrument rather than a retrospective report, turning an unobservable quality risk in distributed human evaluation into a measured, controllable process.

Boundary conditions. The mechanism applies wherever coordination reliability and benchmark validity can diverge and an external benchmark is intermittently observable; clinical second-opinion panels, content moderation, audit review, and professional certification share this structure. Where the benchmark is fully observable, governance reduces to ordinary quality control; where it is never observable, validity is unidentified and only reliability can be managed. The governed regime we study is the intermediate, and common, case in which the benchmark appears often enough to calibrate but not often enough to verify every judgment.

References

[1] Darby, M. R., & Karni, E. (1973). Free competition and the optimal amount of fraud. *The Journal of Law and Economics*, 16(1), 67–88.

[2] Dulleck, U., & Kerschbamer, R. (2006). On doctors, mechanics, and computer specialists: The economics of credence goods. *Journal of Economic Literature*, 44(1), 5–42.

[3] Hayes, R. H., & Pisano, G. P. (1996). Manufacturing strategy: at the intersection of two paradigm shifts. *Production and Operations Management*, 5(1), 25–41.

[4] Lorenz, J., Rauhut, H., Schweitzer, F., & Helbing, D. (2011). How social influence can undermine the wisdom of crowd effect. *Proceedings of the National Academy of Sciences*, 108(22), 9020–9025.

[5] Spearman, C. (1910). Correlation calculated from faulty data. *British Journal of Psychology*, 3(3), 271–295.

[6] Hayes, A. F., & Krippendorff, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures*, 1(1), 77–89.

[7] Lawrence, I., & Lin, K. (1989). A concordance correlation coefficient to evaluate reproducibility. *Biometrics*, 255–268.

[8] Bland, J. M., & Altman, D. G. (2010). Statistical methods for assessing agreement between two methods of clinical measurement. *International Journal of Nursing Studies*, 47(8), 931–936.

[9] Messick, S. (1994). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *ETS Research Report Series*, 1994(2), i–28.

[10] Spearman, C. (1987). The proof and measurement of association between two things. *The American Journal of Psychology*, 100(3/4), 441–471.

[11] Lord, F. M., & Novick, M. R. (Eds.). (2008). *Statistical theories of mental test scores*. IAP.

[12] Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2), 187–204.

[13] Crompton, H., Burke, D., Nickel, C., Bozkurt, A., Miao, F., Sharples, M., Greene, J. A., & Parsons, D. (2026). Governing generative AI in higher education: A global Delphi study on policy and practice. *International Journal of Educational Technology in Higher Education*, 23(1), 21.

[14] Tai, J., Ajjawi, R., Boud, D., Dawson, P., & Panadero, E. (2018). Developing evaluative judgement: Enabling students to make decisions about the quality of work. *Higher Education*, 76(3), 467–481.

[15] Carless, D., & Boud, D. (2018). The development of student feedback literacy: Enabling uptake of feedback. *Assessment & Evaluation in Higher Education*, 43(8), 1315–1325.